

Performing with Me: Enhancing Audience-Performer Interaction in An Immersive Virtual Play

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ABSTRACT

Immersive theater in virtual reality (VR) has been increasingly popular in recently years. However, it is difficult for the audience to immerse in the play while interacting with the performer who usually presented as virtual avatars. In this paper, we present a novel approach for enhancing audience-performer interaction based on the theory of immersive theater. We integrated three types of interaction into a virtual play: individual-based interaction, scenario-based interaction, and narrative-based interaction. We reported the findings of a pilot qualitative study and identified three design recommendations: (1) letting the audience decide how the scenario goes, (2) making clear the identity of the audience, and (3) increasing the audience's freedom to explore in VR space. These recommendations could be adapted for future VR theater experience.

CCS CONCEPTS

• Human-centered computing; • Virtual reality;; • Applied computing; • Arts and humanities; Performing arts.;

KEYWORDS

Immersive theater, Virtual reality, Audience-performer interaction, Motion capture

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1 INTRODUCTION

The relationship between theater and virtual reality technology has been ongoing since the initial development of VR in the 1980s [1]. Immersive theater has become increasingly popular in recent years, which differentiates itself from traditional theater by removing the stage and immersing audiences within the performance itself [2–4]. Virtual reality is an ideal technological medium which engages audience through imaginary environment and interactive narrative [5]. An immersive virtual play usually conducts in a computer-generated world that consists of the real performer and the audience [6]. However, the interaction between the audience and the performer has not been deeply studied. In this paper we propose a novel experience that integrating three interaction types in an immersive virtual play: individual-based interaction, scenario-based interaction, and narrative-based interaction. We consider how the interaction patterns created new opportunities for character identification and the design of narrative structure and the virtual scenarios where the audience would experience in the play.

Immersive theaters in virtual reality. Based on previous works, the definition of VR theater includes both 360-degree panorama of an existing theatrical experience [7–9] and the computer-generated environment [10–13]. The computer-generated environment goes much beyond the traditional theater setting that engage the audience at the nexus of gaming and entertainment. Beyond these approaches, we find a new trend towards integrating real performer with virtual scenarios in immersive theater, which immerses audiences within the performer's self-expressions and the surroundings [14, 15]. For example, Wu et al. propose a system that allows performers to animate virtual characters in real time as they interact with a user in VR [6]. Kaneko et al. present a VR-based remote system that enables audience to exchange non-verbal communication with the performers by using body actions [16]. From a more narrative perspective, Leo et. al investigate the use of physical props in VR to tell the story of an ant's life [17]. Plaxen et al. explore how to position players from a third person perspective where the narrative branches based on the objects they interact with [18]. Gupta et al. describe the design and evaluation

of The Next Fairy Tale VR, a theatrical interactive storytelling system that supported roleplaying for the player [19]. However, these approaches are biased towards realistic scenario and narrative structure [20], the interaction between the audience and the performer has not been well studied.

Interacting with virtual avatars. Researchers have studied human-avatar interactions in virtual reality for several decades [21–23]. Previous studies show utility of human-avatar interactions in aspects of promoting emotional interactions. For example, Kim et al. find there are significant differences in the recognition memory between the initiating and responding interactions in human-avatar communication [24]. Aseeri et al. investigate the influence of avatar representation and behavior on communication by comparing three conditions of avatar appearance [25]. Yakura et al. examine computational approaches to enhance the sense of co-presence through virtual audience avatars by copy user’s movements [26]. Yan et al. create a VR performance which enables asynchronous interaction with virtual avatars [27]. However, these studies concentrate more on natural dialogues or behaviors, for the virtual play, it is important for the audience to smoothly communicate with the performer and immerse in the scenario at the same time. In this paper we explore these possibilities with a new approach to enhance audience-performer interaction. We structure the contributions of the paper in the following manner:

- We construct an immersive virtual play in real-time and augment three types of interaction between the audience and the performer.
- We present the findings of a pilot study of that design focused on identifying interaction features that enhanced the performing experience.
- We propose three design recommendations that can better support immersive theater experiences in VR.

2 DESIGNING THE IMMERSIVE VIRTUAL PLAY “LETTER FROM AN UNKNOWN WOMAN”

The immersive virtual play is created based on the novel “*Letter from an Unknown Woman*” by Zweig as its theme. We recreated it in a drama version and designed the theater performance in VR. This story is mainly about the heroine wrote a long letter in her last moment of life to express her desperate love to a writer. In the play, the audience becomes the last listener of the heroine before her death. Through different types of interaction with the performer and progressive narrative of the story, the audience will be gradually engaged in the plot.

2.1 System Design

We divided the virtual play into three scenarios (“*Whisper of the heart*”, “*First sight in childhood*”, “*Back to Vienna*”) according to the narrative structure. All of them were indoor scenes. The scenes were implemented by using Unity3D. Unity3D is utilized due to its rich 3D resources. Scene, lighting, sounds and special effects as stage machinery entities are modeled and constructed for use in scene generation. We created both stage scenarios and the virtual avatar of the performer. We used Noitom™ Perception Neuron 2.0 to capture the performer’s body movements and used iFacialMocap plug to

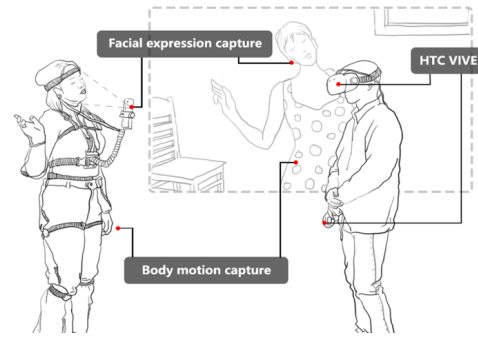


Figure 1: The system of the immersive virtual play.

capture facial expression. The play was displayed on HTC VIVE, and the corresponding processor was a 4GHz computer with 16GB RAM and NVIDIA GeForce GTX 980Ti. HTC VIVE has a resolution of 1200 by 1080 pixels per eye and weighing of 0.55kg. The system enabled performer to observe the virtual scenes from audience’s view, which could help her to adjust the position. We set up the experiment in a 20 square meter testing place of our laboratory, the avatar which represented the performer would interact with the audience in three scenarios in real-time. The system design was shown as Figure 1.

2.2 Audience-Performer Interaction

Our main goal in this play was to create an immersive virtual space that provided audience different types of interaction with the real performer and the freedom to explore in virtual scenes. We proposed three behavioral augmentations for audience-performer interaction (see Figure 2). In the following we describe the design on each type of the interaction in detail.

2.2.1 Individual-based Interaction (IBI). The expression of individual opinions, emotions or comments is the key factor affecting interactive experience for the audience [28]. Individual-based Interaction (IBI) which is augmented with natural communications (such as free talk and physical contact). The IBI appears six times in both “*Whisper of the heart*” and “*Back to Vienna*”. “*Whisper of the heart*” is the first scenario of the play. The residence of the performer’s avatar is displayed in the scene. In “*Back to Vienna*”, the performer’s avatar will tell the experiences of her reunion with Mr. R. The IBI mainly includes language interaction and physical contact: language interaction refers to the natural dialogues between the audience and the performer based on the plot. For example, the performer will ask the audience about their identity and intention, whether they are willing to listen to her own stories, or whether they have the same experience. Physical contact is reflected at the time when the story is about to end. In “*Back to Vienna*”, the performer will ask the audience whether they are willing to give her a hug. If the audience agrees, she will go and hug them. Physical contact only appears once at the end of the play, but it is the audience’s choice to decide at which scenario it is going to take place.

2.2.2 Scenario-based Interaction (SBI). In an immersive theater, it is important for audience to perceive their surroundings [13, 29].



Figure 2: (a) The IBI in third-person view, (b) the IBI in VR mode; (c) the SBI in third-person view, (d) the SBI in VR mode; (e) the NBI in third-person view, (f) the NBI in VR mode.

Drawing on this idea, we propose scenario-based interaction (SBI) that enables audience to play directly with virtual stage props under the guidance of the performer. The SBI appears five times in all scenarios. For example, in *"Whisper of the heart"*, the performer will ask the audience to turn on the light at the beginning of the play. The audience will touch the switch on the wall with the hand controller, then the scene turns bright. In addition, the performer's avatar will ask the audience to help her hand over a letter at the end of the story. With the guidance from the performer, the audience will pick up the letter and read it with the performer. In *"First sight in childhood"*, the performer will instruct the audience to pick up the phone. When the audience touches the microphone with the hand controller, the audio of relevant contents is triggered on and the audience will be instantly teleported to another place to watch a video. The audience can touch the virtual door behind them under the prompts of the subtitles at the end of the video and return to the previous position. In *"Back to Vienna"*, the performer's avatar will instruct the audience to pull the trigger, using their hand controllers, to lay the roses scattered on the table into the vase. During the process, the volunteer will spray perfume around the audience so that they can also obtain the pleased sense of smell.

2.2.3 Narrative-based Interaction (NBI). Narrative-based Interaction which is augmented with multiple story line. The NBI only appears once in *"Back to Vienna"*. The performer's avatar will provide two options for the audience when communicating with them, and the audience's option will determine the final of the story.

When the performer's avatar states that she has been forgotten by Mr. R again, she will ask the audience whether she will "have a quarrel" or "leave quietly". If the audience chooses the former, the performer will continue to act in the scenario according to the audience's option. The ending of the story is that she leaves the city with her child after having a fierce quarrel at Mr. R's. and her child dies of influenza later. If the audience chooses the latter, the performer will act according to the first scenario *"Whisper of the heart"*. The ending of the story then is that she quietly leaves Mr. R's, chooses to end her life after her child dies of influenza and asks the audience to hand over her last letter to Mr. R before she dies.

3 METHOD

Three females and three males were recruited for the experiment. They were all undergraduates or postgraduates majored in design and aged between 18 and 24. Before the experiment, they were asked to fill in the forms of basic information and the acknowledgement of VR. The experiment was conducted as follows.

The whole play lasted about 20 minutes. After the deduction of the multiple line ending, the duration of *"Whisper of the heart"*, *"First sight in childhood"* and *"Back to Vienna"* was about 3 minutes, 5 minutes, and 7 minutes respectively. And that of the branch-line endings of the two scenarios of *"Whisper of the heart"* and *"Back to Vienna"* were 5 minutes and 4 minutes respectively. According to the changes of the performer's position in the virtual space, the audiences turned their bodies to adjust the viewing angles and

switched between different scenarios. After the experiment, each of the six audiences was being semi-structured interviewed for an average duration of about 20 minutes. They were asked about their understandings and feelings of the play. There were 13 questions in the interview, which were classified into three dimensions: immersion experience, narrative experience, and interaction experience, with 5 questions, 4 questions and 4 questions respectively. The interview process was recorded in writing and audio forms. As a result, four significant findings that affecting the interaction between the audience and the performer were summed up, and three specific design recommendations were formed after discussion.

4 FINDINGS

4.1 Connecting with the narrative

There were five audience in total who thought that the NBI was their favorite, which enabled them to participate in the play. Performer adjusted herself according to the audience's choice and gave corresponding feedback, letting the audience to be dominant in the plot. At the same time, the audience were able to make choice for the character freely based on the context of the current performance content. This could attract audience's attention effectively and made them understand the plot deeply.

P02 Among all interaction types, what impressed me most was that the heroine asked me to help her make choice. I chose "a big fight", because I thought Mr.R's behaviors were intolerant. After I made my choice, she told me that she broke the vases of Mr.R and left. I thought I made the best choice for her.

P04 I couldn't understand the heroine's love for Mr.R, so for most of the time, I did not see the performer but observed the environment all the time. However, when the heroine asked me to help her, she attracted my attention again. This behavior made me understand more what the story talked about.

4.2 The importance of identification and role-playing

We found that making the identity of the audience clear could bring them positive experience. Two thirds of the audience hoped that they could have an identity that fitting the plot and could get some prompts and guidance to strengthen their understanding of the character identity at the same time. Otherwise, the audience would doubt over the role they played in the plot, making them unknown about how to respond to the performers in what identity and with what attitude.

P06 I felt that I was out of the story. Sometimes I couldn't realize that the performer was talking to me. When the performer asked me to help her make choice, I thought that I had no responsibility to be involved in the private affairs. If I could be given an identity in advance, I would feel more immersed.

P01 I hoped we could make the role of the audience clearer, because the person kept changing when the performer was telling a story. Sometimes it was "you" and sometimes it was "him", which made me confused. I did

not know if I should make choice on behalf of Mr. R or as a spectator.

4.3 Factors that enhanced interactive experience

We found that the uneven distribution of interaction frequency and interaction type in the virtual play would affect the audience's experience. Almost all the audience thought that there should be more interactions during the entire performance. In addition, there were three audience said that the difference of the interaction type among the three scenarios would also affect their experience.

P01 I thought that the interval between every two interactions was too long, which affected my feeling of expecting and exciting. There were only few interactions, so it was difficult for me to focus on the play.

P05 I hoped that there could be more interactions when we entered the scenario, so I could be more immersed. It was like a newbie guide for games, which made me more familiar with the specific operations and flows.

P06 thought that the distribution of all interaction types was uneven among the three scenarios. He observed that language interaction and physical contact were presented in the first and the last scenario, and there was only scenario-based interaction in "First sight in childhood". Narrative-based interaction only appeared once. This distribution made him unsatisfied. He thought that if we could distribute different interaction types combined with the plot evenly into different scenarios, the experience would be better.

4.4 Freedom in VR performance space

We should take the audience's needs to explore the performance space independently into account. In this play, the movement area was limited. The audience could turn their bodies to adjust the viewing angles and switched between different scenarios. The interactions could be completed in the area that were close enough for them to achieve. Some audience thought that their behaviors were limited in the space. P05 thought that if the audience could actively explore the space and trigger the plot, they would be more immersed.

P05 Compared with being told that "the performance has begun", I wished more to walk around by myself as an explorer to find out the plot. For example, following the sound, I found a performer crying in the corner. When I was at her side, she realized me and started to communicate with me. I liked this better.

P06 thought that all interactions were center around the plot, promoting the narration with strong purpose. However, some implicit interactions were very necessary. Such interactions might not require guidance and prompt from the performer, and the audience could find out these implicit clues through freely exploring in the scenario.

5 DESIGN RECOMMENDATIONS

5.1 Letting the audience decide how the scenario goes

Our first design recommendation is to let the audience decide how the scenario goes. We suggest increasing the frequency of narrative-based interaction, which means that it will appear 2 to 3 times in each of the three scenarios, making the audience maintain continuity and continuous decision-making behaviors throughout the whole narrative line. Therefore, they can realize that they are important during the whole experience and are more participated and focused on the plot. The key node of each decision should be provided by the performer and naturally blended into the plot and there should be certain correlation between the former and the latter decisions. For example, in the scenario of “*First sight in childhood*”, the audience can help the heroine choose whether to go to Mr. R’s again before she leaves. Audience’s choice decides whether the heroine will commit suicide at the end of the plot. Such setting let the audience realize that each of their choice is very important. Method to prompt a decision can be the dialogue from the performer or scenario texts. For example, in “*Whisper of the heart*”, the texts can be set in the scenario to let the audience choose whether to close the door before leaving. Similarly, the decision of the audience will decide whether the heroine will commit suicide successfully.

5.2 Making clear the identity of the audience

Our second design recommendation is to make the identity of the audience clear. We suggest that such prompts should be mainly distributed in the scenarios when the play just begins to strengthen the identity of the audience and let them adapt to their identity as fast as possible, paving the way for the development of the follow-up plot. We suggest the performer that giving repeated and multiple prompts to let the audience not miss the sign of identity confirmation. At the same time, such sign should also be subtly blended in the plot so as not to affect the performing experience. For individual-based interaction, the performer’s tone will be decided by the audience’s identity. For scenario-based interaction, for example, in “*Whisper of the heart*”, the audience can visit the heroine as “a little girl lives next door”, which makes the heroine remind when she was young as maid, then she starts to tell her stories to the audience. Such setting makes the plot more reasonable. Scenario elements featuring reflection effect, such as mirror, plate, can be added in “*First sight in childhood*” and “*Back to Vienna*”. Audience can perceive their appearance presented on the reflected side, which enhances the sense of identification.

5.3 Increasing audience’s freedom to explore in VR space

Our third design recommendation is to increase audience’s freedom to explore in VR space. We suggest distributing interactive elements in all scenarios, rather than limit to certain specific area. The performer can encourage the audience to explore freely while they are performing. For example, in “*Whisper of the heart*”, the heroine can invite audience to “have a look freely” in her home and at the same time narrate her story. In “*Back to Vienna*”, the heroine

talked about the experience of delivering a baby, we can put the photos mentioned in the play as interactive clues. The audience can view the photos to deepen the understanding of the plot. When the audience is exploring freely, they should avoid missing key plots. At this time, guidance of light and sound can be adopted to inform the audience the area that they can explore. The performer also needs to adjust their position and lines in the space. If the audience is distracted by other interactive elements, the performer can walk to the audience or attract the audience again through language interaction. Without influencing the story line, we should offer rich sensual experience as far as possible to let the audience feel that they are focused by the performer all the time, triggering empathy.

6 CONCLUSION AND FUTURE WORK

In this paper, we integrated three different types of interaction between the audience and the performer in an immersive virtual play. The three types of interaction are individual-based interaction, scenario-based interaction, and narrative-based interaction. We recruited six audiences and ran a small qualitative interview pilot study of audience experiencing this virtual play to better understand how the different types of audience-performer interaction could enhance the immersive experience. In the end, we summarized three design recommendations: (1) letting the audience decide how the scenario goes, (2) making clear the identity of the audience, and (3) increasing the audience’s freedom to explore in VR space.

In the future, we plan to conduct deeper research in the following three aspects: (1) improving the quality of scenario design and the virtual avatar for a delicate VR theater experience, (2) collaborating with theater professionals to apply the prototype in practical applications, (3) evaluating audience experience by integrating behavior study and biometric measurements.

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